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SURFACE ALBEDO TREATMENT SYSTEM – DEFLECTING NEAR EARTH ASTEROIDS WITH PAINT

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Much research and techniques have been proposed for the purpose of deflecting near earth asteroids (NEAs) which can hit the Earth creating great destruction. The project outlined here which has been worked on for the last seven years takes advantage of altering the Yarkovsky effect, a phenomenon due to uneven thermal irradiation on a surface which creates a noticeable force on small rotating bodies in the solar system. Using a modified industrial powder gun to spray paint the surface of an asteroid with powder (otherwise known as albedo change particles), the albedo (reflectivity) of the asteroid can be modified which changes the amount of photons emitted on each side of the surface. By focusing on the particular asteroid Apophis, we designed an entire mission through simulations and ground experiments which uses Apophis as the test case scenario for our surface albedo treatment system (SATS) payload. We go into details of particular experimental design parameters, experimental setup, positions and velocities of albedo change particles (ACPs), albedo change observed, and the uncertainty of the asteroid before and after albedo change, we construct a coherent picture of the entirety of the design of our SATS payload.

INTRODUCTION

Much research and techniques have been proposed for deflecting near earth asteroids (NEAs) which have the probability of impacting the Earth resulting in catastrophic damage. ECAPS LLC proposes an innovative slow push technique using a payload on a spacecraft known as the surface albedo treatment system (SATS). The albedo modification is implemented by painting the surface of the asteroid with albedo changing powders (ACPs) via a triboelectric dispenser, hence directly changing the Yarkovsky effect. To put it simply, the technique is to paint an asteroid surface and change its color to change its trajectory.[1]

The Yarkovsky effect is a force due to the anisotropic thermal distribution on a rotating body which generates a force imbalance between the dawn side and the dusk side as the thermal photons leave the surface. The amount of light absorbed depends on the albedo, i.e. the reflectivity of the surface. This effect is particularly prevalent among small bodies between 10 cm and 10 km. The effect is strong size-dependent but even for a large kilometer scale asteroid, the steady push of the Yarkovsky effect can be enough to transport it vast distances.

The design and implementation of SATS follows a three-step process. The first step is an initial study of three parts discussed in this paper called the Apophis Mitigation Technology (AMT). The second step is a low-earth orbit (LEO) flight test of a spacecraft housing the SATS system with anticipated launch in 2018. The final step is an asteroid exploration and mitigation mission called the Apophis Exploration and Mitigation Platform (AEMP) to the near earth asteroid Apophis with anticipated launch in 2021 and arrival at Apophis in 2022. With almost zero probability of hitting the Earth, Apophis's close approach in 2029 offers an excellent opportunity to explore and test the long-term effects of the mitigation technique.

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The design and implementation of the Apophis Exploration and Mitigation Platform is discussed in the prior mission outline papers[1].

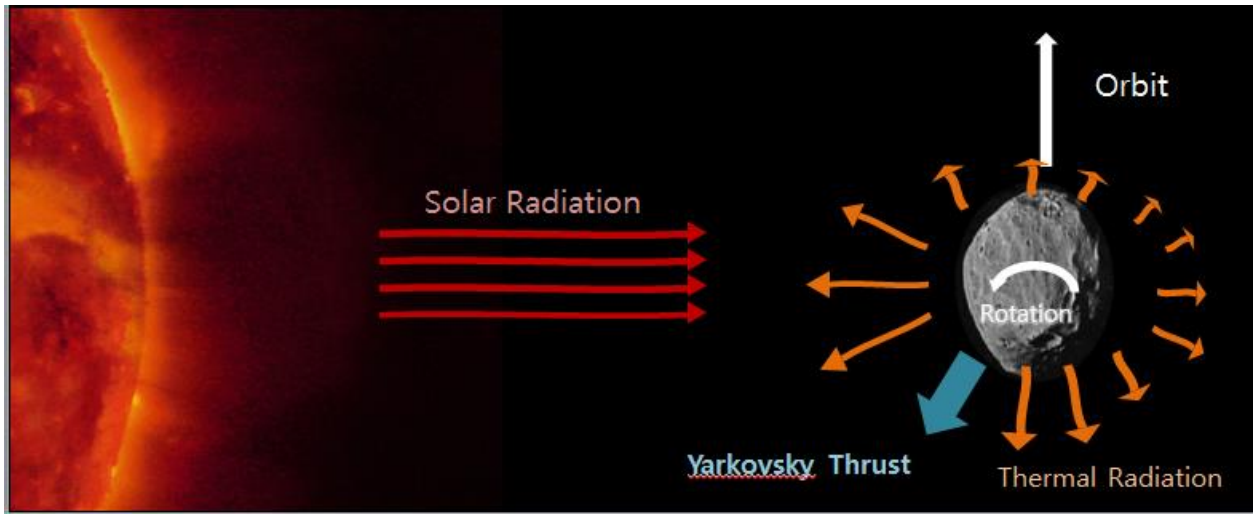

Figure 1. Daylight side absorbs solar radiation. As object rotates, dusk side emits more light than dawn.[3]

GROUND EXPERIMENT

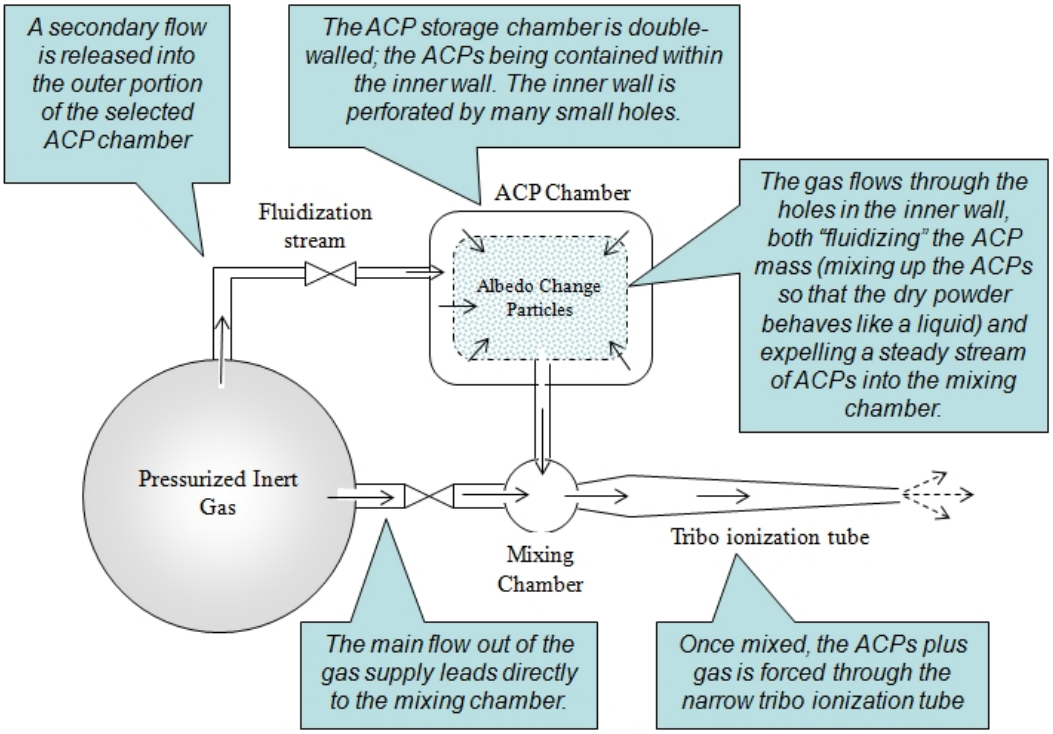

Figure 2. Schematic depiction of the triboelectric dispenser mechanism.

The SATS is composed of four main components as shown in Figure 2: inert gas chamber, ACP chamber, mixing chamber, and triboionization tube[2]. The core design requires modifying a powder painting gun used on Earth commonly for industrial powder coating, which is commonly used for painting car rims, furniture, door knobs, and a variety of other terrestrial objects. A modified Nordson

Corporation power painting tribodispenser is been developed for use in space. The operation of the SATS in the space environment follows such steps,

1. Pressurized inert gas forces power through a narrow passageway, where the dispenser wall material is an electric donor and the ACPs are electron acceptors.
2. Dispenser releases the dry, negatively charged ACPs onto the positively charged asteroid surface.
3. Solar radiation melts and cures the powder, bonding the particles firmly to each other and the surface.

The ground experiment combined with simulations discussed later serves to optimize a set of parameters. These four parameters in order of importance are as follows,

1. Albedo change on surface (AC) is the amount of total change in color and reflectivity on the surface which is ultimate factor of how much the Yarkovsky effect will be affected.
2. Average charge/mass ratio of particles (Q/M) is the average charge-to-mass ratio of the particles immediately at the end of the tube.
3. First pass transfer efficiency (FPTE) is the ratio of the mass of powder on the surface to the mass of power ejected.
4. Coverage mass versus area covered ratio (A/M) is the ratio of the area of surface to the mass of the powder on the surface.

Optimizing this set of parameters requires developing an appropriate ACP and an optimal SATS setup. For the former, the ACP must have good physical and chemical characteristics. Physically, the particles must be greater than 100 microns in diameter so that they will not levitate due to dust elevation effect. Chemically, the particles must have a combination of factors,

- High solar absorptivity at near-IR wavelengths to ensure melting before curing on the surface of the asteroid. The powder must be able to melt at 100 – 120 C.
- Low light intensity to cross-link to ensure successful curing after melting on the surface of the asteroid.
- High electronegativity to ensure successful charge accumulation while the particles are traveling through the tube.

According to the Powder Coating Research Group (PCRG), the combination of factors is currently unfulfilled by any powder. PCRG can develop a new powder to develop these and other issues. The SATS setup requires a complex interplay of interconnected design parameters as seen in Table 1. To reduce the number of experiments, an experimental testing optimization technique called Taguchi's method can be used.

Table 1. Independent-dependent variable correlations.

COMPONENT	Q/M	AC	A/M	FPTE
Tube Length	X		X	X
Tube Material	X			
Tube Radius	X		X	X
Tube Nozzle	X	X	X	X
Gas Pressures	X	X	X	X
Particle/Gas Ratio	X	X	X	X
Surface Albedo		X		

The ground test differs from the LEO experiment with the presence of gravity and on a smaller scale. Every effort is implemented to simulate space conditions including vacuum and sunlight. A vacuum chamber has been built and an UV projector models sunlight. The apparatus is composed of the

pressurized gas source, fluidized bed (hopper), a hose connecting the hopper to the mixing chamber, main ionization tube, and nozzle. A schematic of the setup can be seen in Figure 3.

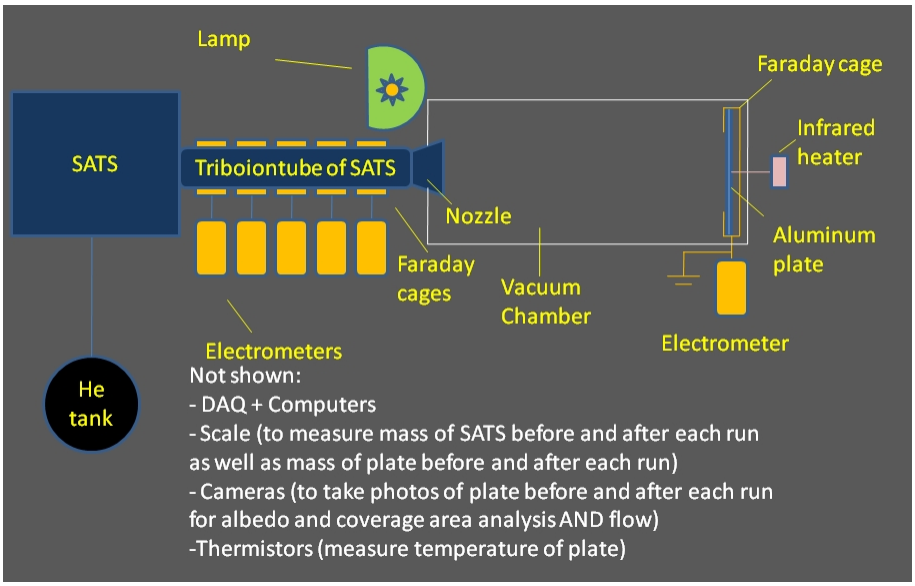

Figure 3. Schematic of the ground experiment for optimizing design of SATS.[2]

A vacuum chamber has been recently built and tested for operations. The design procedure included two main steps: 1) an approximation of the time required to evacuate the chamber with the vacuum pump that had already been purchased, and 2) a calculation of the maximum amount of time that tribodispenser could be discharged into the chamber before the desired level of vacuum was lost. These calculations led to the idea of using an additional volume connected in-line in the system to increase our maximum spray time by increasing the total vacuum volume. Figure 4 shows the vacuum chamber and respective apparatuses including vacuum gage, vacuum pump, and vacuum pump fittings.

Figure 4. Vacuum chamber without the gasket (left), vacuum gauge (middle), vacuum pump (right top), and vacuum pump fitting (right bottom).

Initially, the ACPs will land on a charged flat uniform aluminum surface. The four parameters of interest will be examined after a series of experiments with varying parameters. Later, a mock-up asteroid surface will be used. For all experiments, cameras capture the before and after scene of the test surface.

SIMULATIONS

The ACP Dynamic Simulator (ADS) requires a variety of target asteroids to be simulated for each design.[2] By testing against a variety of targets, more likely scenarios can be weighted accordingly and limiting cases can be found. Particle flow within and near the dispenser have been simulated via Monte Carlo techniques, combined with computational fluid dynamics of the transporting gas, and iterated upon to improve estimates of the charge distribution and its resulting force on the particles. Preliminary studies have yield promising results but there remains the work in defining the user defined functions for the flow characteristics.

Investigations of the dispenser dynamics begin with a series of laboratory experiments. The goal of these experiments is to verify the models developed for design and analysis. These models include, but are not limited to

- Particle charge density
- Mass flow rate
- Spray pattern and velocity

The simulator study so far has operated on a number of design and target scenarios [1]. However, more scenarios and target cases must be examined for more rigorous validation. The effect on the albedo and other values such as efficiency are to be further quantified. By comparing the results of design iterations over the target asteroids, we can find optimal regions in the design space. The design variables are as follows:

- Mass flow rate
- Average particle size
- Average particle charge
- Spray pattern and velocity (nozzle design)
- Dispensing altitude

The asteroid variables are:

- Surface charge density
- Shadow pattern
- Mass density
- Radius

Note that the initial simulation will assume a spherical target asteroid. Future versions will be able to account for irregular shapes. Each particle will be subject to a normal distribution in size, charge, and initial conditions with respect to the particular case's design parameters. The number of particles simulated will be large enough to provide a representative distribution for the design.

Fluid Dynamics Simulations

The gas flow is considered as incompressible, steady and turbulent.[2] Since the mass and momentum loadings of the ACP are high, the interaction between the continuous phase (gas) and the ACP is included in the model to account for the impact of the solid phase on the flow field of the gas phase. The effect of particle volume fraction on the gas flow is neglected as the volumetric fraction of the solid phase is very low.

The tribo dispenser tube is assumed to be of constant diameter to spray the ACPs into space without the aid of a converging or diverging nozzle. The electric force interactions between the ACPs will cause them to repel each other and spread out into a spray cone as they travel. After the ACPs are injected through the nozzle, it will be dominated by the near-field dynamics. In this near-field region, we consider both the influence of the ACP/gas flow field and electrostatic field. The trajectories of the particles are anticipated by solving the particle motion equation. In each *i*- direction, the velocity and position of the particle at given time can be calculated.

To calculate the electrostatic force (F_E), we need several parameters, which are the particle charge (q_p,), the displacement (a) between the particle and the asteroid, and the unit vector (n_p) from the particle to the surface of the asteroid with minimum value of a. The inputs for the model are all parameters to solve the primary forces: the drag force, the gravitational force, the electrostatic force. The density of the particle is used to calculate the drag force. The particle mass, particle charge are needed for the electrostatic force. Consequently, we solve the position and velocity of the particle as the outputs.

The charging model used calculates the charge is based on the quantity of each particle's collisions with the walls and their velocities at the time of the collisions.[5] The alpha parameter quantifies the charge buildup due to rolling and slipping; it can only be determined experimentally. The charging model is depicted in Figure 5.

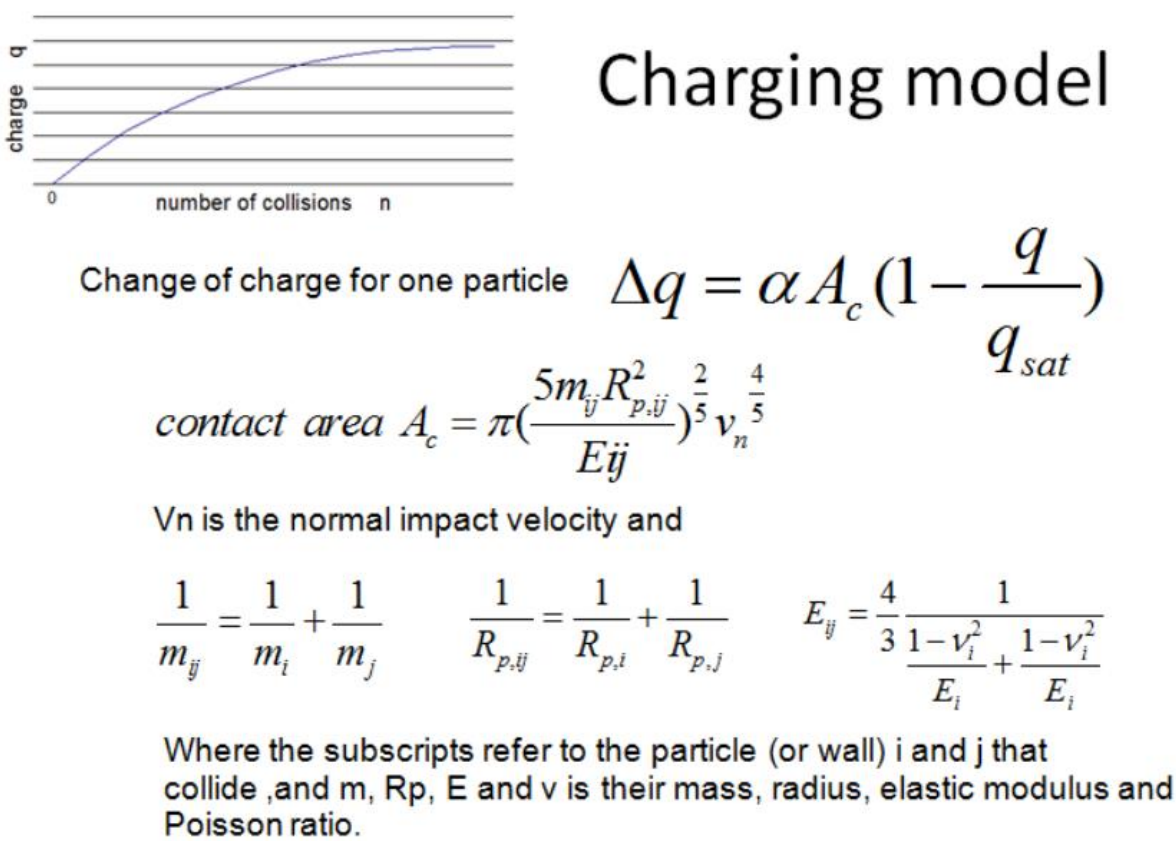

Figure 5. Charging equations for modeling particle collisions.

Initially, the model was made in SolidWorks and was then imported into OpenFOAM. Within OpenFOAM, the solver used pimpleCoupledKinematicParcelFoam.[2] This solver combines the basic solver pimpleFoam and icoUncoupledKinematicParcelFoam together to simulate transient progress of the

movement for both flow and particles. PimpleFoam is a solver in OpenFOAM for the calculation of flow based on PISO and SIMPLE algorithm. IcoUncoupledKinematicParcelFoam is a solver in OpenFOAM for the calculation of particles based on discrete element method, which includes the interaction between particles and both other particles and walls. Particle flow through a cylinder was simulated in OpenFOAM as seen in Figure 6.

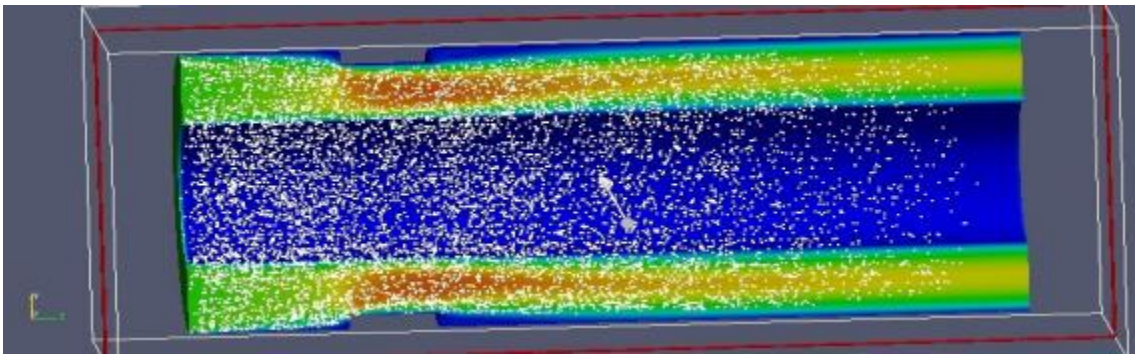

Figure 6. Particle movement in the flow without charging through a cylinder in OpenFOAM.

Unfortunately, OpenFOAM is very good at computational fluid dynamics (CFD) but is poor at particle simulation.[5] Furthermore, OpenFOAM's particle simulation is insufficient for the task of charge modeling. For this reason, an external program written in MATLAB/Octave called the ChargeBuildProgram is used to accurately model both particle velocities and charges. The solution algorithm is depicted in Figure 7.

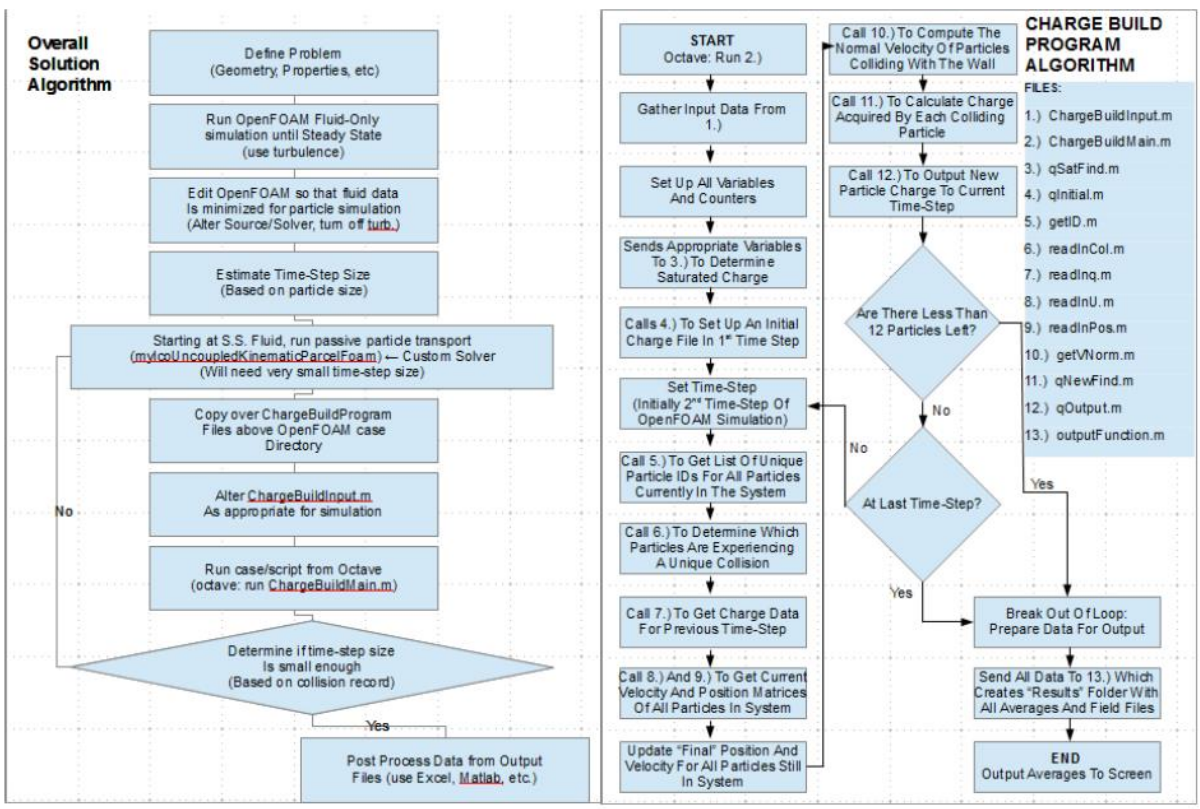

Figure 7. Solution algorithm of the fluid dynamics portion of the simulation.

Simulation results with particles indicate that the final velocities of the particle at the end of the tube follows a double hump in distribution for simulation time steps of 1 microsecond and 0.5 microsecond.[7] The distribution is under investigation. The magnitude of velocity distributions are depicted in Figure 8.

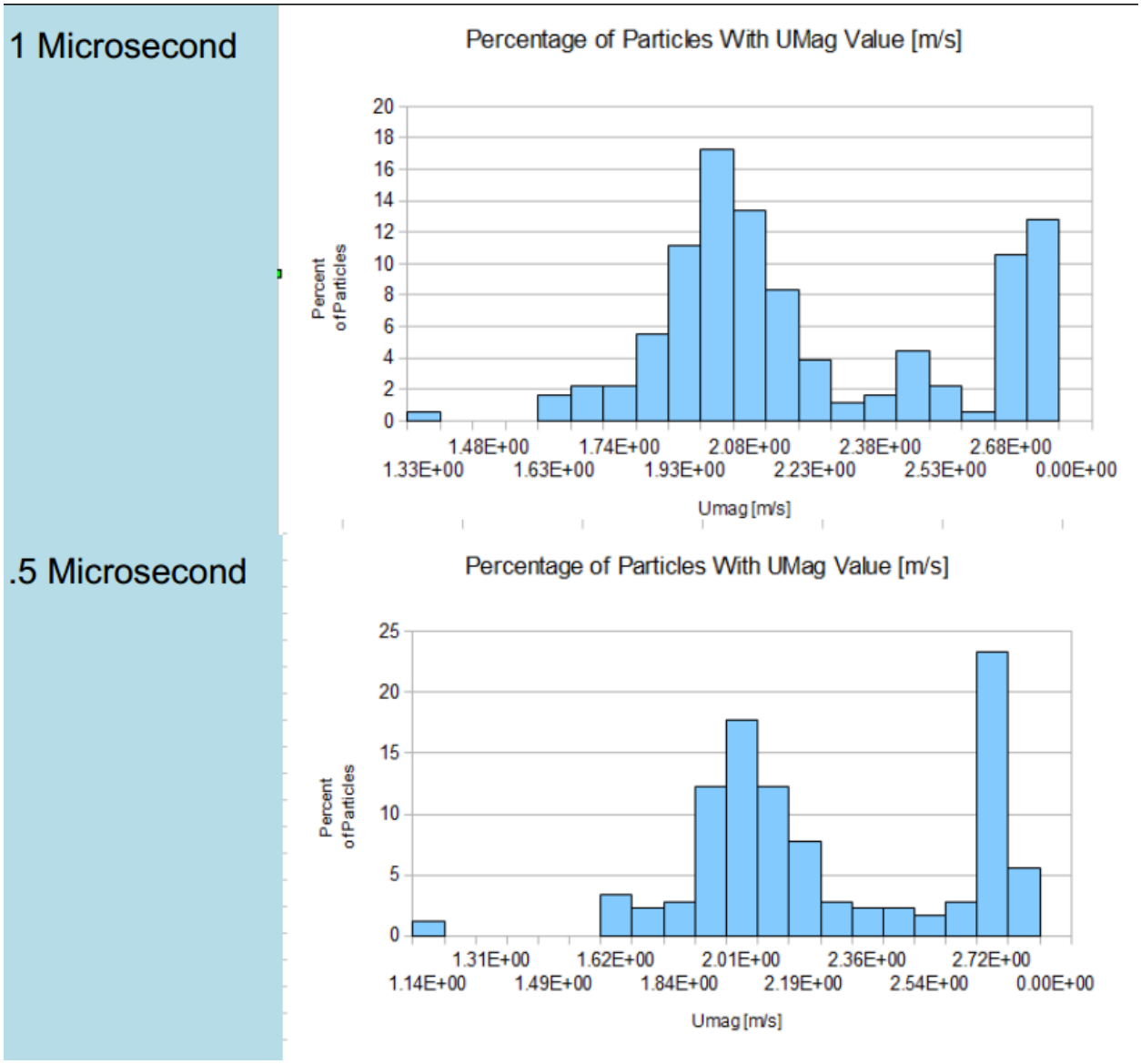

Figure 8. Velocity magnitude of the particles at the end of the tribodispenser tube.

For both sets of time steps, various alpha values in the charging model are used to determine the effect on particle charging. The actual alpha values can only be determined by measuring data from actual ground experiments and fitting the data to the simulations. Preliminary estimates are shown in Figure 9.

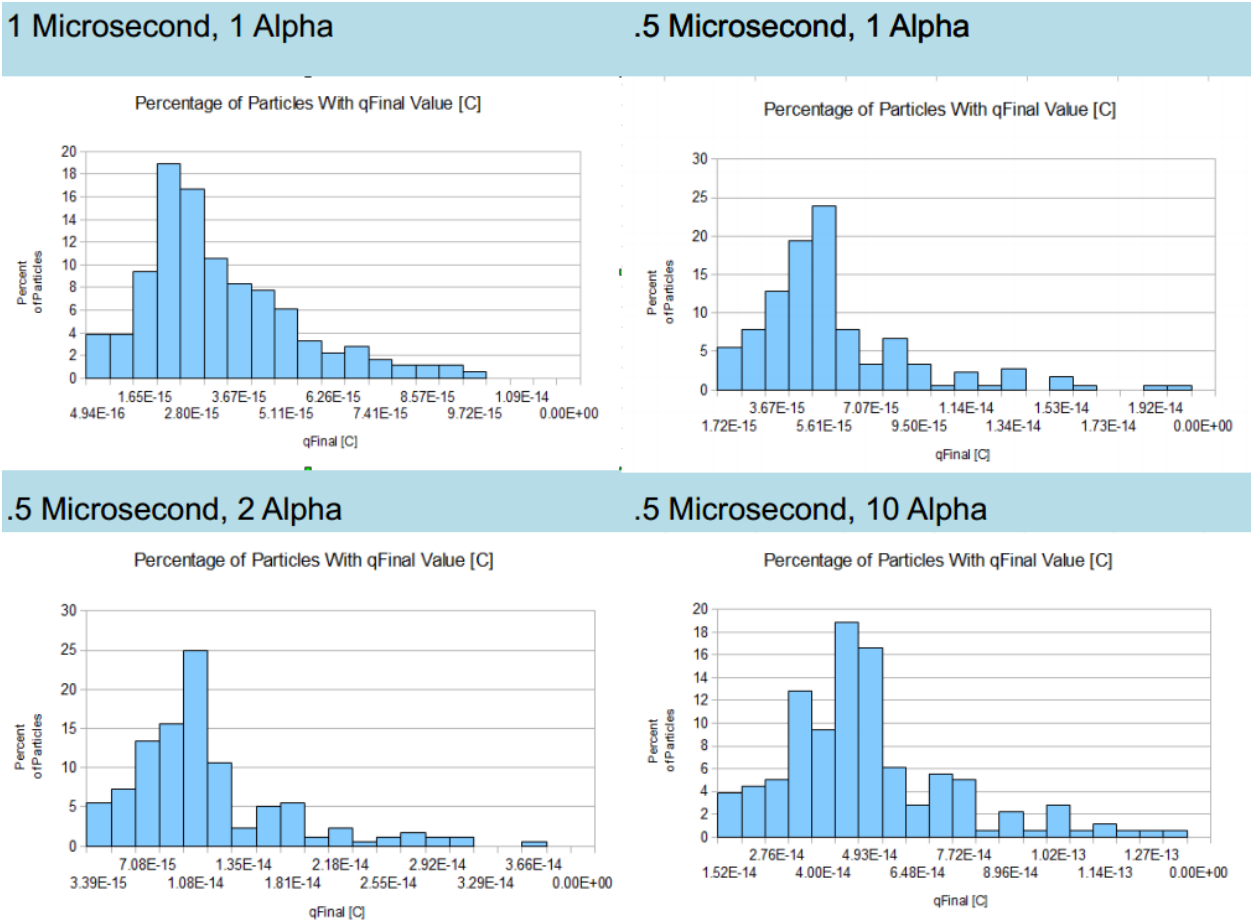

Figure 9. Charge of the particles at the end of the tribodispenser tube with different assumed alphas.

Inter-space Simulations

To simulate the ACPs path from the tribo-dispenser to the asteroid surface, four operations must be performed. First, each particle must be assigned a time at which it will leave the tribo-dispenser and what its states are at that time. Next, the various environmental effects that exert forces on each ACP are computed at each time-step. Third, at each time-step the forces are taken into account to propagate the trajectory of each ACP via numerical integration. Finally, each ACP is checked for collision with the asteroid at each time-step.

The operations on each ACP are independent of every other ACP, i.e. the gravitational force on the ith ACP is independent of the gravitational force on the jth ACP. For this reason, the computation of forces, trajectory propagation, and collision detection can all be performed in parallel for each ACP. Since there are many thousands of ACPs, the algorithms can be parallelized to take advantage of thousands of processors as is the case in a GPU. To use the GPU for these computations, a framework was constructed to allow for some parts of the simulation on the CPU while also having some parts on the GPU. The GPU promises massive parallelization at the cost of slow data transfer and difficulty in the program development. The CPU allows for easy data storage and manipulation at the cost of minimal parallelization. The simulation framework aims to balance these costs and benefits. The implementation of the GPU yielded several orders of magnitude decrease in computation time when compared to a pure CPU implementation.

To model the collisions between the ACPs and the asteroid, a high resolution map of the asteroids surface is needed. This map must be able to be referenced efficiently at every time-step to determine if an

ACP has collided and if so where. The primary difficulty in this task is handling the irregular shape of an asteroid. The method employed here does not completely capture a completely arbitrary shape for the asteroid; however, the results have proved to be accurate for the model of asteroid Itokawa.

Six height/normal maps are generated by projecting the asteroid's surface onto both sides of the three orthogonal coordinate system planes. These maps contain the distance of the asteroid's surface from the plane and the surface's normal map. These maps can be visualized as a 2D image with pixel values from black to white indicating the projected distances, Figure 10 and Figure 11.

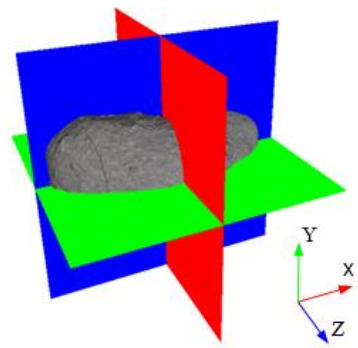

Figure 10. Orthogonal collision map orientations

Figure 11. Example height maps for the XY plane. The surface on the negative-z side of the XY plane is on the left. The positive-z surface is on the right. The color denotes the distance of the surface from the XY plane with black as zero and white as 200 meters.

Currently, the simulation includes a randomized generation of particles at the end of the tribo-dispenser nozzle based on a uniform probability distribution that creates a cone-like spray field. Once in flight, the particles currently interact due to gravity and electrostatic forces. The mass and charge of the particles can easily be updated via a GUI as better ACP data becomes available.

The ACP particles and other bodies with mass interact based on the universal law of gravitation,

$$\mathbf{F} = G\frac{M_1 M_2}{R^2}\hat{R} \tag{1}$$

The ACP particles are each given a charge when created. Currently, all particles have the same charge although a probability distribution would be preferable. Based on the charge, a force is computed based on Coulomb's law,

$$\mathbf{F} = k\frac{Q_i Q_{Ap}}{R^2}\hat{R} \tag{2}$$

At every time-step, the force on each ACP is computed. These forces will induce accelerations on each particle which can be integrated numerically to determine the ACPs' positions and velocities. The numerical integration scheme implemented in the simulation is the Adams-Bashforth method. The Adam-Bashforth method is a multistep method for numerical integration. Compared to the Runge-Kutta method, the Adam-Bashforth method is computationally more efficient. The Runge-Kutta method evaluates function derivatives an times interior to the time-step which is expensive and disregards the function values at all times prior to the current time. The Adams-Bashforth method does not require interior derivatives and uses past function values to achieve order-of-accuracy equivalent to those of the Runge-Kutta method. The first through fourth order Adams-Bashforth methods have the forms

$$\begin{aligned}
y_{n+1} &= y_n + h\left(f\left(t_n, y_n\right)\right) \\
y_{n+2} &= y_{n+1} + h\left(\frac{3}{2} f\left(t_{n+1}, y_{n+1}\right) - \frac{1}{2} f\left(t_n, y_n\right)\right) \\
y_{n+3} &= y_{n+2} + h\left(\frac{23}{12} f\left(t_{n+2}, y_{n+2}\right) - \frac{4}{3} f\left(t_{n+1}, y_{n+1}\right) + \frac{5}{12} f\left(t_n, y_n\right)\right) \\
y_{n+4} &= y_{n+3} + h\left(\frac{55}{24} f\left(t_{n+3}, y_{n+3}\right) - \frac{59}{24} f\left(t_{n+2}, y_{n+2}\right) + \frac{37}{24} f\left(t_{n+1}, y_{n+1}\right) - \frac{3}{8} f\left(t_n, y_n\right)\right)
\end{aligned} \tag{3}$$

Currently, the first order method is implemented; however, provisions are in place to let the user select which order of accuracy to use based on their needs and computational cost. Additionally, the user can specify the time-step.

The question of whether the gravitational effect of bodies far away should be considered has been analyzed, namely the Sun, Earth, Venus, Mars, and Jupiter. Every particle will feel approximately the same gravitational attraction to these objects. Apophis, the spacecraft, and the particles will all be attracted with nearly the same force as if they were a rigid body. More properly, the *spatial gradient* of the gravitational attraction to a body far away is extremely low.

The acceleration of a particle is given by

$$a = G\frac{M_{Sol}}{R^2}$$

which has the spatial gradient

$$\frac{da}{dR} = -2G\frac{M_{Sol}}{R^3}\ .$$

Consider the position of two particles as R and $R + \Delta R$ respectively. Their accelerations are thus

$$a_1 = G\frac{M_{Sol}}{R^2}, a_2 = G\frac{M_{Sol}}{R^2} + \frac{da}{dR}\Delta R\ .$$

The difference between their accelerations is $\frac{da}{dR}\Delta R$. If ΔR is on the order of a meter, and R is about an astronomical unit, the difference between the accelerations is

$$\Delta a = a_2 - a_1 = -2G\frac{M_{Sol}}{R^3}\Delta R = -2GM_{Sol}O\left(1e-33\right)$$

In order to optimize the performance of the tribo-dispenser system, an array of metrics need to be defined to quantitatively compare the results of several trials with different parameters. The metrics aim

to describe the development of the distribution of ACPs as they fly from the tribo-dispenser to the asteroid's surface. The distribution of ACPs can be described several ways, but the simplest and most informative method is to record the position of each ACP as it passes through several planes on its way to the asteroid surface as depicted in Figure 12.

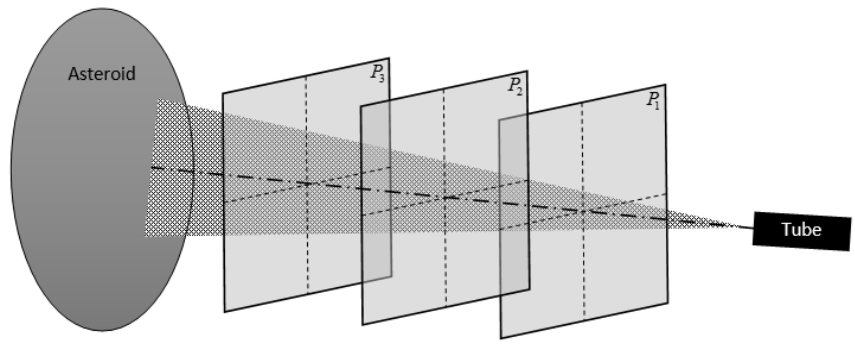

Figure 12. Schematic of the planes used to quantify the ACP distribution development.

Assuming the center axis of the tribo-dispenser tube lies on the z-axis, the radial position of the ith particle on the jth plane is computed as

$$R_{i,j} = \sqrt{x_i^2 + y_i^2} \ .$$

A histogram sorting operation then determines the number of ACPs at several radius intervals. For the data here the intervals are two meters wide and go up to 50 meters, i.e.

$$r_k = 2.5k, \ k = 0..20$$

The number of particles in the radial interval r_{k-1} to r_k is denoted as n_k for $k = 1..20$. The discrete probability distribution is thus $p(r_k) = \dfrac{n_k}{N\pi\left(r_k^2 - r_{k-1}^2\right)}, \ k = 1..20$

A sample distribution is shown in Figure 13. Shown in the probability distribution are the data for a sample case along with a fit to the probability distribution. The best fit for each jth plane is a uniform distribution up to a transition radius for the jth plane $r_{t,j}$ after which the distribution is a one sided Gaussian. The transition radius grows linear as the particles travel away from the tribo-dispenser, and the Gaussian standard deviation grows as a square of the distance.

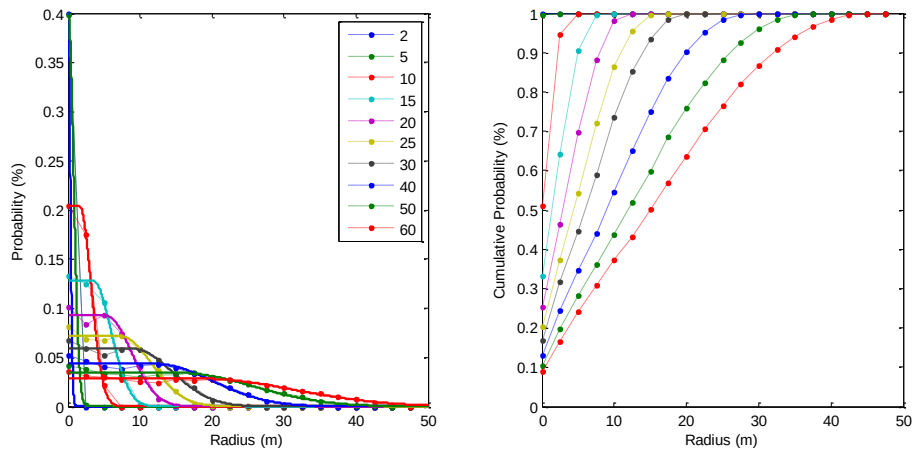

Figure 13. Sample probability distribution and cumulative probability distributions for a sample simulation. Each line represents the distribution at a plane at a specified distance from the tribo-dispenser tube.

Trajectory Change Simulations

The trajectory change required to deflect an asteroid such as Apophis needs to be calculated to determine the effectiveness of the technique. The code starts with 2D and 3D normal distribution of positions and velocities with mean and covariance values. The 6D sphere of uncertainty will be propagated from Apophis's initial conditions to 2036 with Monte Carlo simulations. A schematic of such propagation is in Figure 14.

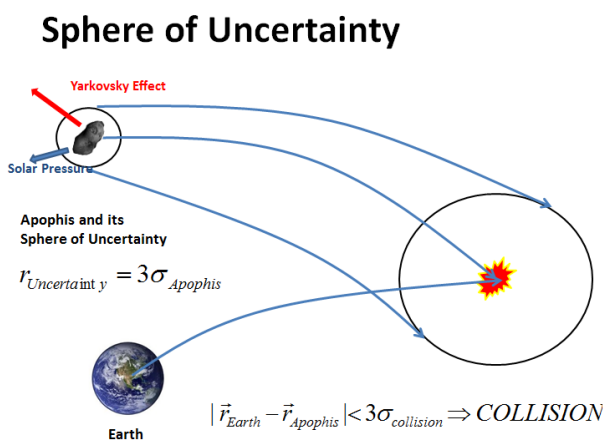

Figure 14. Schematic of Trajectory Propagation of Apophis

Without considering Earth's gravitational pull, the collision criteria is defined as

$$|\vec{r}_{Earth} - \vec{r}_{Apophis}| < 3\sigma_{collision} \Rightarrow COLLISION$$

whereas by considering with the Earth gravitational pull which is our case, the equation is

$$|(\vec{r}_{Earth} + \vec{R}_{Influence}) - \vec{r}_{Apophis}| < 3\sigma_{collision} \Rightarrow COLLISION$$

This propagation off trajectories requires an accurate dynamical model and a powerful integrator.

1. Dynamic Model: an N-body problem was considered including Sun, and all eight planets plus Pluto. Also Earth's Moon was added to the system of equations and three big asteroids by the names Ceres, Pallas, and Vesta. The set of ordinary differential equations to be solved has 90 states (fifteen objects and six states each). Also the solar pressure and Eath J2 effect is included for the case of close Apophis-Earth encounter.

2. Integrator: seveal integrators were tested and their results were checked with the reults of NASA-JPL HORIZONS website for validation. MATLAB standard integrator ode45, along with Runge-Kutta 10th order (with 17 steps), and Adam-Bashfoth Adam-Moulton (with different orders) were tested. The technique Adam-Bashforth Adam-Moulton 5th oder yielded the best performance in Apophis trajectory propagation. We refer to this method as ABM5 for the sake simplicity. ABM5 uses a predictor-corrector (explicit-implicit) in the heart of its algoithm. Table 2 depicts some resuls produced by ABM5 mehod versus those of NASA/JPL. The initial condition was considered Oct.2, 2013 00:00:00 CT and final times are from Oct.2, 2014 to Apr. 2, 2029 00:00:00 CT. The close-encounter was predicted as Friday Apr.13, 2029 which is in agreement with NASA/JPL HORIZONS website. The second and third columns are the results (distance between Apophis and Earth $\delta_f(km)$) using ABM and NASA/JPL and the last column shows the relative percantage error defined as *error%=|true-estimated|/true.*

Table 2. Earth-Apophis Distance in km based on ABM5 and JPL website.

Final Time	ABM5 $\delta_f(km)$	JPL $\delta_f(km)$	Error (%)
Oct.2, 2014	209862939.210922	209862807.137569	0.000063%
Oct.2, 2018	252120230.435658	252120813.444156	0.00023%
Oct.2, 2023	285560493.048409	285560753.459342	0.000091%
Oct.2, 2028	44997110.0392671	44989956.3409284	0.016%
Dec.2, 2028	63937184.6653079	63928557.9189882	0.013%
Feb.2, 2029	39955283.7723685	39950434.4898995	0.012%
Apr.2, 2029	6122900.76824476	6122674.50139698	0.0037%

Using the integrator ABM5, uncertainty propagation can now be performed for the three phases namely, pre close-encounter, through close-encounter, and post close-encounter. Figure 15 shows the intial spehere of uncertainty as of Oct.2, 2013 and Table 3 presents the growth of the sphere of uncertainty for the three phases. Since the close encountr occurs on Apr.13, 2029, we consider all dates before Apr.2, 2029 as pre, Apr.2 through May2. 2029 as through, and after May2, 2029 as post close-encounter periods.

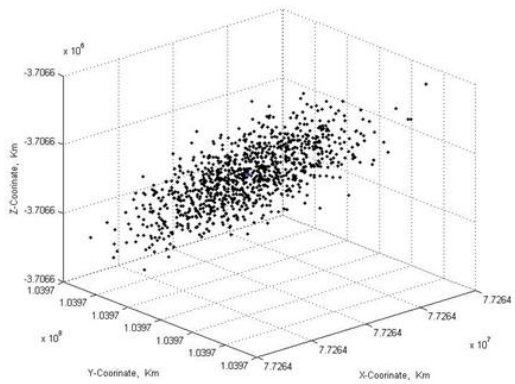

Figure 15. Initial Spehere of Uncertainty as of Oct.2, 2013.

Table 3. Sphere of Uncertainty Growth vs. Time.

Sphere Radius	Oct.2, 2013	Apr.2, 2029	May2, 2029	Oct.2, 2036
$3\sigma_r$	155 km	900 km	1,011 km	1,760 km

As can be seen from Table 3, the radius of uncertainty sphere ($3\sigma_r$) grows almost 11.4 times, but since the distance between the earth and Apophis is in the order of millions of kilometers in the year 2036, the uncertainty associated wih the initial condition on Oct.2, 2013 cannot be a source of concern.

To have a more accurate trajectory and uncertainty propagation, Yarkovsky effect also shoulkd be added to the dynamical model. In other words, we want to calculate how much powder on Apophis do we need to change Apophis's trajectory sufficiently enough to not hit Earth in 2036.

The acceleration produced by Yarkovsky efect has two components namely,

1) Diurnal acceleration which is dealing with rotation of the asteroid (in oder of hours) and is perpendicular to the asteroid spin axis $(\mathbf{a}_d.\mathbf{s}=0)$ where $\mathbf{a}_d$ is the diurnal Yarkovsky acceleration and $\mathbf{s}$ is the asteroid spin unit vector

2) Seasonal acceleration which is dealing with the transition of the asteroid (in oder of years) and is parallel to the asteroid spin axis, $\mathbf{a}_s = f_z\mathbf{s}$, where $\mathbf{a}_s$ is the seasonal Yarkovsky acceleration vector and f_z is the acceleration magnitude.

SPACECRAFT DESIGN

The selected preliminary design, called the Static Preliminary Albedo Demonstration Experiment (SPADE), illustrated in Figure 16, involves a non-deploying, target surface rigidly fixed to the satellite. To meet mission requirements, the orbit must be no more than 700km, to avoid excess radiation, and no less than 300km to meet the minimal mission lifetime of 30 days. An inclination of at least 31° is required for communications with a TAMU ground station. The configuration is set to be at an altitude from Earth surface of 350 km, an eccentricity of 0°, and an inclination of 46° leading to an orbital period of 91.5 minutes.[3] The mission duration is for 30 days and 4 tests will be conducted. The spacecraft mass is estimated to be 58.51 kg with 16.23 kg attributed to the structural mass.

The tribo-dispenser is to be mounted at the top of the spacecraft, with the barrel angled down towards the surface.[4] The flow of the particles will spread out in a cone with a circular cross-section, finally curing and thus adhering to the surface. The particles will be stored in a spherical tank near the tribo-dispenser, with pressurant stored in a separate tank. Upon release, the negatively charged particles will be electrostatically attracted to the surface, where they will cure. Images will be captured during deposition and curing. These images will be stored and forwarded to the ground station. Experiment observation will be performed by a monochrome camera.[4] Color imagery is unnecessary to measure the change in visible light intensity, increases the data load and decreases sensitivity. The sun will provide the light for imaging.

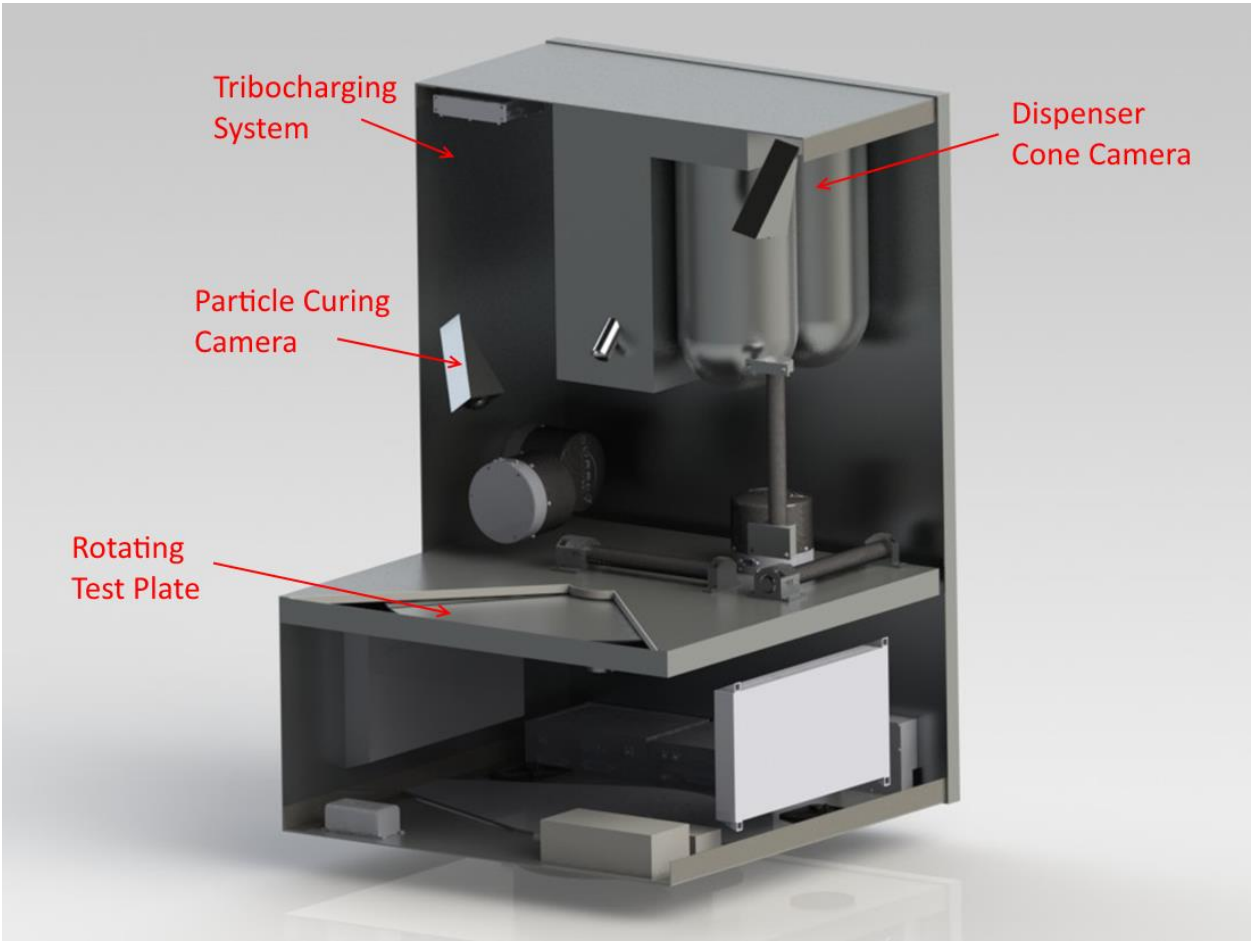

Figure 16. Refined configuration of the AMT spacecraft bus.[3]

The SPADE mission is designed as a secondary payload. For the purposes of this design the launch vehicle was assumed to be the Minotaur IV or an equivalent launch vehicle, capable of using an ESPA 24 payload adapter. The maximum size of a spacecraft attached to an ESPA 24 adapter ring is 24 in x 20 in x 18 in. The ESPA 24 payload adapter ring uses a lightband separation mechanism to release satellites from the adaptor.[4]

The mission requires that while the experiment is being performed, the test surface is normal to the sun and trailing in the spacecraft's plasma wake. Spacecraft attitude will be controlled with reaction wheels with three magnetic torque rods as backups. This allows 3-axes controllability. The guidance hardware of the spacecraft will use sun sensors and magnetometers. The former is desired due to its simplicity and low power while the latter is required due to the presence of magnetic torque rods. The attitude control system concept is summarized in Figure 17.

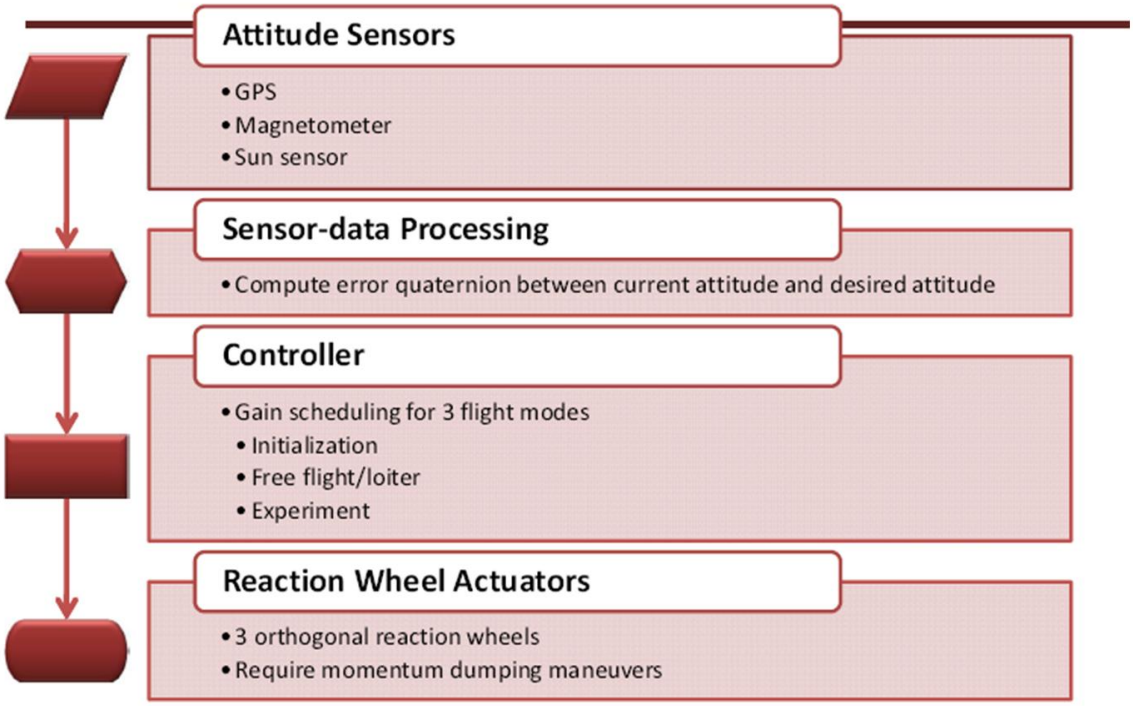

Figure 17. Step-by-step process to sense and control the LEO spacecraft.[3]

The telecommunications system consists of two SST S-Band Patch Antennas, a SST S-Band 4 Watt Downlink Transmitter, and a SST S-Band Receiver. The 8.0 mbps data rate for transmission requires 30.7 minutes of communication time for experimental images to downlink to the ground station. With 210 depositions per test, 90 curings per test, and 2 scratch observations per test, 1208 pictures taken in total for four tests at a resolution of 1032 x 776 pixels are required. The 9.6 kbps uplink data rate is sufficient for experimental run and stop commands. Notably, due to the inclination chosen, the average view duration is only 2.8 minutes with two average views per hour. Multiple ground stations are required due to the low communication time and window size. The total time in view is 133 minutes for as can be seen in results generated shown in Figure 18.

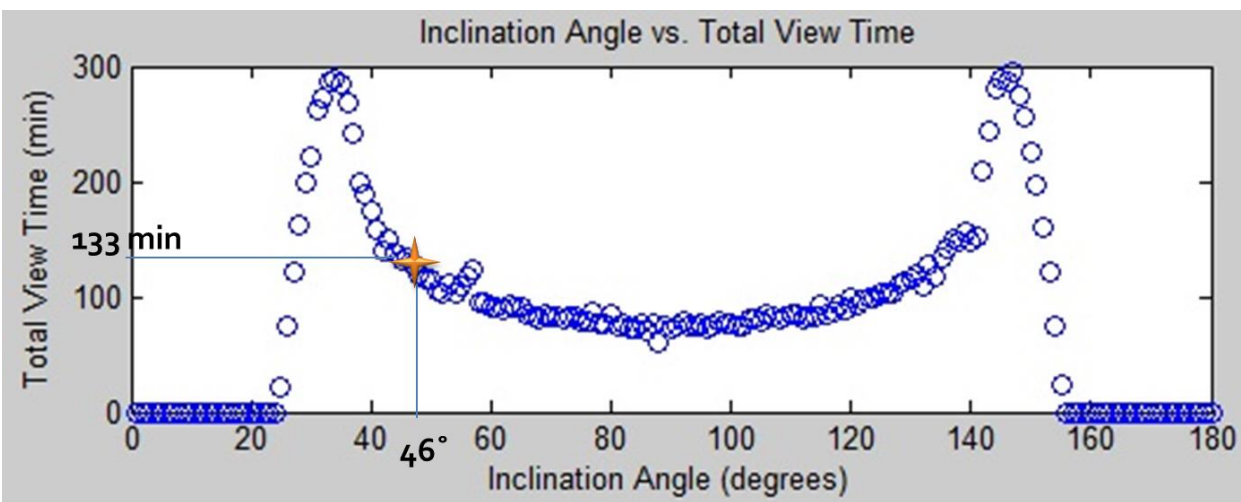

Figure 18. Low communication time and window size suggest the need for multiple ground stations.[3]

Power for the satellite is divided into payload power and satellite power. The experimental payload requires approximately 100 W for the duration of the experiment, provided by a lithium thionyl chloride (Li-SOCl2) primary battery.[4] The tribo-dispenser will require approximately 100 W of power for the duration of the application process. A dedicated primary battery will be used to supply power to the tribo-dispenser, while solar panels with a secondary battery will support the bus. The bus requires a continuous 20 W. This power is to be provided by body-mounted gallium arsenide (GaAs) solar panels paired with a nickel cadmium (NiCd) secondary battery. Thermal analysis shows that the worst-case hot temperature for the satellite is 305 K, while the worst-case cold temperature is 230 K, when using a passive thermal control system.

A model of the LEO spacecraft constructed from welded metal plates can be seen in Figure 19.

Figure 19. A mockup of the LEO spacecraft made from metal plates.

CONCLUSION

In this paper, the focus has been on the AMT project of ground experiments, simulations, and LEO spacecraft design. This work precludes the second layer of the LEO flight experiment and the final layer of the AEMP to the actual asteroid Apophis. The SATS system in all three layers as can be seen in Figure 18 provides many desirable characteristics for changing the trajectory of a potentially threatening

asteroid. A tribodispenser design uses techniques that are commonly used and well understood. The SATS systems can continually later the orbit of Apophis or similar NEAs far beyond a mission lifetime and eventually it can eliminate the threat of impact.

With the effectiveness of the trajectory change observed via the test mission to Apophis, the application method can then be applied to any NEAs given a sufficient warning time. In the case of a certain impact and enough warning time, a mitigation mission using SATS can drastically reduce design time and mitigation cost.

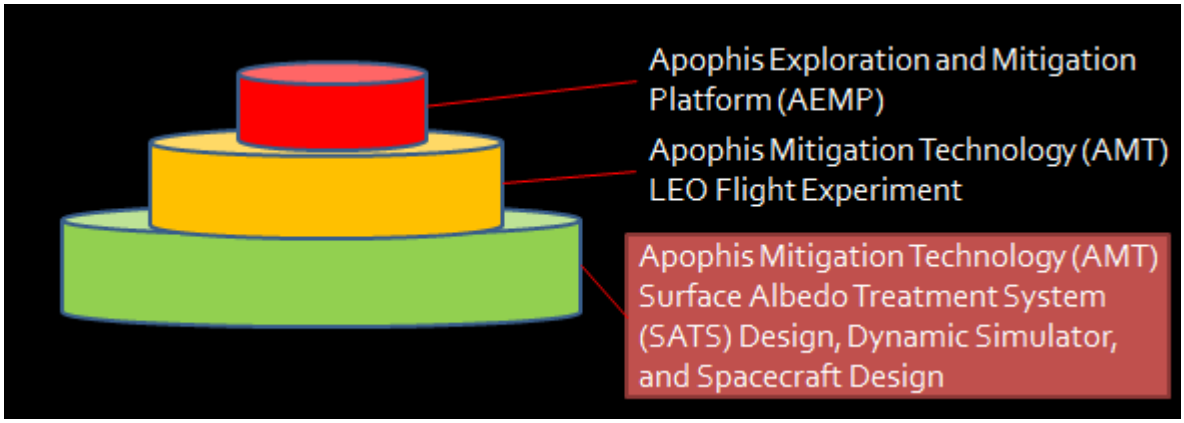

Figure 20. Detailed three step process for exploration and mitigation near earth asteroids with SATS.[3]

NOTATION

ACP	Albedo Change Particle
AEMP	Apophis Exploration and Mitigation Platform
AMT	Apophis Mitigation Technology
LEO	Low Earth Orbit
NEA	Near Earth Asteroid
PCRG	Powder Coating Research Group
SATS	Surface Albedo Treatment System
SPADE	Static Preliminary Albedo Demonstration Experiment

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